

Consider dividing

$$10x^2 + 4x + 1$$

by

$$5x - 3$$

in $\mathbb{Z}_{11}[x]$.

$$\bullet \quad 5x - 3 \equiv 5x + 8 \pmod{11}$$

Long division:

$$\begin{array}{r} 2x + 2 \quad R \ 7 \\ 5x + 8 \overline{) 10x^2 + 4x + 1} \\ \underline{-(10x^2 + 5x)} \quad (16 \equiv 5 \pmod{11}) \\ 10x + 1 \quad (-1 \equiv 10 \pmod{11}) \\ \underline{-(10x + 5)} \\ 7 \quad (-4 \equiv 7 \pmod{11}) \end{array}$$

THUS

$$10x^2 + 4x + 1 \equiv (5x + 8)(2x^2 + 2) + 7 \pmod{11}$$

quotient

remainder.